

Hello Typst!

Table of contents

0.1 Parameters	1
0.2 Shearrestistance	1
0.3 Function	1
0.4 Conditional	1

0.1 Parameters

$$\begin{aligned} \mathcal{O}_{sw} &= 12\text{mm} & s_{sw} &= 150\text{mm} & z &= 190\text{mm} & V_{Rd}^{s1} &= V_{Rd}^s \left(10\text{mm}, 150\text{mm}, \frac{435\text{N}}{\text{mm}^2}, 200\text{mm}, 25^\circ \right) = 48.85\text{kN} \\ f_{sd} &= 435.0 \frac{\text{N}}{\text{mm}^2} & \alpha &= 25^\circ & V_{Rd}^{s2} &= V_{Rd}^s \left(10\text{mm}, 150\text{mm}, \frac{435\text{N}}{\text{mm}^2}, 120\text{mm}, 45^\circ \right) = 29.31\text{kN} \end{aligned}$$

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$$\begin{aligned} & \backslash \mathrm{mm} \quad s_{sw} \\ & = 150 \mathrm{mm} \quad \backslash \mathrm{mm} \\ z &= 190 \mathrm{mm} \quad \backslash \mathrm{mm} \quad \& \quad f_{sd} \\ & = 435.0 \frac{\mathrm{N}}{\mathrm{mm}^2} \\ \alpha &= 25^\circ \end{aligned}$$


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0.4 Conditional

$$z = \begin{cases} \sqrt{300}\text{mm} & \text{for } \alpha \geq 25^\circ \\ \ln(200\text{mm}) & \text{for } \alpha \geq 10^\circ \\ 500\text{mm} & \text{otherwise} \end{cases} = 17.32\text{mm}$$

$$b_w = 100\text{mm} \quad f_{cd} = 20.0 \frac{\text{N}}{\text{mm}^2}$$

0.2 Shearrestistance

$$a_{sw} = \frac{\mathcal{O}_{sw}^2 \pi}{4s_{sw}} = 753.98 \frac{\text{mm}^2}{\text{m}}$$

$$V_{Rd}^s = \frac{zf_{sd}a_{sw}}{\tan(\alpha)} = 133.64\text{kN}$$

$$A_s = \frac{\mathcal{O}_{sw}^2 \pi}{4} = 113.1\text{mm}^2$$

$$x = \frac{A_s f_{sd}}{0.85b_w f_{cd}} = 28.94\text{mm}$$

0.3 Function

the function body is indented

$$V_{Rd}^s(\mathcal{O}_{sw}, s_{sw}, f_s, z, \alpha) = \frac{zf_s a_s}{\tan(\alpha)}$$

$$a_s = \frac{\mathcal{O}_{sw}^2 \pi}{4s_{sw}}$$

$$\alpha = \begin{cases} 43^\circ & \text{for } a_s \geq \frac{20\text{mm}^2}{\text{m}} + \frac{\mathcal{O}_{sw}\text{mm}^2}{\text{m}} \\ 22^\circ & \text{for } a_s \geq \frac{30\text{mm}^2}{\text{m}} \\ \alpha & \text{otherwise} \end{cases} = 25^\circ$$